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A DATA ENVELOPMENT MODEL FOR AGGREGATING PREFERENCE RANKINGS*

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This paper presents a general model for aggregating votes from a preferential ballot. The thrust of the model is to accord each candidate a fair assessment in terms of his overall standing vis-a-vis first place, second place, . . . , kth place votes. The form of the model is a combined index $\sum_{j=1}^{k} W_j v_{ij}$ where v_{ij} is the number of the jth place votes received by the ith candidate. The weights W_j are assumed to form a monotonically decreasing sequence with $W_j - W_{j+1} \geq d(j, \epsilon)$. These constraints correspond to the assurance region (AR) side constraints in the DEA framework. The properties of the model are examined in terms of this discrimination intensity function d, and in the special case that $d(j, \epsilon) = \epsilon$, our model is shown to be equivalent to the consensus models of Borda and Kendall.
(VOTING; RANKING; DEA; CONSENSUS)

1. Introduction

In a preferential election each voter selects a subset of k candidates from a ballot of m choices, and rank orders these k candidates from most to least preferred. Such a voting format is common in municipal elections, where a number of candidates are required to fill various positions. This same structure appears in other prioritization settings, such as the ranking of projects, products in a consumer survey, etc. We shall utilize the preferential voting example for discussion purposes throughout the paper.

A problem of interest for over 200 years has to do with the *aggregation* of votes from preferential ballots. Borda (1781) proposed the "Method of Marks" as a means of deriving a consensus of opinions. This method amounts to determining the average of the ranks assigned by voters to each candidate, with the winning candidate being the one with the lowest average. An equivalent version of this model was later presented by Kendall (1962). Cook and Seiford (1982) have extended the Kendall model using an l^2 distance approach. Other distance based models have been advanced by Armstrong et al. (1977), Blin (1976), Cook and Seiford (1978), Cook and Kress (1984), Kemeny and Snell (1962), and others. In these distance models the voter ranks *all* of the alternatives or candidates.

Other models for aggregating preferential votes have arisen from parliamentary settings. The so-called American system, English system and West Australian system are examples of such models, as discussed in Keesey (1974). In some of these models the winning (first place) candidate is determined purely in terms of the maximum number v_{i1} of the first place votes. Here, v_{ij} denotes the number of jth place votes earned by candidate i. *Ties* are often broken (among candidates with the same number of first place votes) by utilizing 2nd place votes, 3rd place votes, etc. Under the American system, for example, the nominee having the least number of first choices is dropped. Those ballots on which that candidate was ranked first are now destroyed. The second choice votes received by the remaining candidates are now added to the first choices received by each. A nominee is declared elected if the total of his first and second place votes constitutes a majority—

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otherwise the candidate with the fewest number of first and second place votes is dropped, and the process is repeated by bringing third place votes into consideration.

Existing preferential voting models are arguably deficient in that they fail to provide a *fair overall* assessment of a candidate's standing—a composite or aggregate evaluation of his first place, second place, third place standings, etc. The problem is one of how to combine the jth place votes, for $j = 1, 2, \ldots, k$, in some rational manner.

In §2 a model is presented which utilizes a *composite index* $Z_i = \sum_{j=1}^{k} W_j v_{ij}$ of the jth place standings of candidate i. The objective of the model is to derive multipliers W_j (the level of importance attached to jth place votes) which will accord a candidate a *fair* assessment of his standing. This is accomplished by allowing for flexibility in the assignment of weights from one candidate to another. Since this process may result in several candidates being tied for first place, the model also provides for maximum *discrimination* among such candidates. This *robustness* property is a principal feature of the model.

The procedure used to derive the W_j, hence the index Z_i, is analogous to the Data Envelopment Analysis (DEA) method of Charnes, Cooper and Rhodes (1978). The DEA method was originally applied to the problem of evaluating the relative efficiency of a set of decision making units (e.g. schools, hospitals, bank branches, courts, etc.).

The usual DEA model possesses two important characteristics which could prove undesirable in the preference ranking setting. First, it is necessary to solve k problems; one for each decision making unit. Second, a number of units will commonly end up being tied for first place, i.e. will be on the efficient frontier. In §3 it is shown that the special properties of our composite index model eliminates both of these characteristics. Specifically, it is shown that only 1 problem, not k problems, needs to be solved to determine a winning candidate. Moreover, it is demonstrated that in most cases only one candidate will end up in first place. Hence, ties for first place will generally be broken, and an ordinal ranking of the k best candidates emerges. It is also shown that a certain special case of this model is equivalent to Borda's method of marks for deriving a consensus among a set of ordinal rankings.

Examples and geometric interpretations are provided.

2. A Fair Model for Aggregating Preferential Votes

In the preferential voting framework each candidate $i = 1, 2, \ldots, m$ receives some number v_{i1} of first place votes, v_{i2} of second place votes, $\ldots$, v_{ik} of kth place votes. The problem is to utilize these votes in a reasonable manner to obtain an overall *desirability index* Z_i for each candidate.

For any given set of weights or multipliers W_j, we define the desirability index for candidate i by

$$Z_i = \sum_{j=1}^{k} W_j v_{ij}. \tag{2.1}$$

Clearly, any preset values W_j are likely to favor some candidates while discriminating against others. In Borda's model, for example, the W_j are the numbers, $1, 2, 3, \ldots, n$. Moreover, the weights are the same for each candidate. What is required is a set of multipliers which provides the *fairest* possible treatment for each candidate. Specifically, we wish to determine a set of multipliers $\{W_j^{i_0}\}$ for candidate i_0 which maximizes Z_{i_0}. To achieve this, we solve the problem

$$Z_{i_0}^* = \max \sum_{j=1}^{k} W_j v_{i_0 j}$$

$$\text{Subject to} \qquad \{W_j\} \in \Phi, \tag{2.2}$$

where Φ is some subset of $\mathcal{R}^k$.

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In the context of preferential voting, the *feasibility* set Φ should be characterized by at least two types of constraints. First, Z_{i_0} should be bounded above, i.e. $Z_{i_0} \leq \theta$ for some θ. Without such a parameter θ, problem (2.2) would be unbounded. Moreover, it is necessary to define some *best attainable* performance level (θ) that any candidate can achieve. As a convention, we set $\theta = 1$ or 100%. As will become apparent, the final rank ordering of the candidates is independent of the choice of θ.

The second set of constraints has to do with the priority attached to the jth versus $(j + 1)$ place votes. It is clear that any reasonable aggregation scheme should be constrained by $W_j > W_{j+1}$. More generally, we define a function $d(j, \epsilon)$: $d: N \times R^+ \rightarrow R^+$, i.e. $d(j, \epsilon)$ is a nonnegative function defined on the cartesian product of the space of positive integers $N \equiv \{1, 2, \ldots, K\}$ and the positive real line R^+. Moreover, $d(j, \epsilon)$ is restricted to be nondecreasing in ϵ. We impose the constraints $W_j - W_{j+1} \geq d(j, \epsilon)$. This lower limit $d(j, \epsilon)$ on the gap between the importance attached to the jth versus $(j + 1)$st place standing is referred to as the *discrimination intensity function*. The parameter ϵ is called the *discriminating factor*.

Problem (2.2) then becomes:

$$Z_{i_0}(\epsilon) = \text{Maximize} \sum_{j=1}^{k} W_j v_{i_0 j}$$

$$\text{Subject to:} \quad \sum_{j=1}^{k} W_j v_{ij} \leq 1, \qquad i = 1, 2, \cdots m,$$

$$W_j - W_{j+1} \geq d(j, \epsilon), \qquad j = 1, 2, \ldots, k-1, \tag{2.3}$$

$$W_k \geq d(k, \epsilon).$$

This problem is now solved for each candidate $i_0 = 1, 2, \ldots, m$. Only the objective function changes each time.

Note that by defining a single "input" variable W_I and input data quantities $U_{iI} = 1$, $i = 1, \ldots, m$, problem (2.3) is equivalent to the well known DEA-AR model. See Thompson et al. (1986, 1989). The constraints $W_j - W_{j+1} \geq d(j, \epsilon)$ represent the assurance region (AR).

To illustrate this problem, consider the case of four candidates, where two of these are to be elected from a preferential ballot. Let the 1st and 2nd place standings be given as follows:

	# Votes	
Candidate i	v_{i1}	v_{i2}
1	6	8
2	4	11
3	8	2
4	3	0

Figure 1 shows the positioning of the four candidates in the $(V_{.1}, V_{.2})$ space. These are referred to as R_1, R_2, R_3, R_4.

If problem (2.3) is now solved for each of the four candidates, the constraint space appears as in Figure 2. In this diagram we have set $d(j, \epsilon) \equiv 0$. Candidate #1 has an optimum either at point A ($W_1 = \frac{3}{34}$, $W_2 = \frac{2}{34}$) or point B ($W_1 = \frac{3}{26}$, $W_2 = \frac{1}{26}$). In any event, the rating for candidate #1 is $Z_1^* = 1.0$. Similarly, candidate #2 has its optimum at the point A where $Z_2^* = 1.0$, and candidate #3 at point B where $Z_3^* = 1.0$. Candidate #4 has its optimum at point C, where $Z_4^* = \frac{3}{8}$.

Referring back to Figure 1, if the points representing candidates #1 and #2 (i.e. $R1$ & $R2$) are joined, the resulting line segment has a slope given by the ratio of the W_1, W_2 coordinates of point A in Figure 2. We refer to the set of line segments joining points

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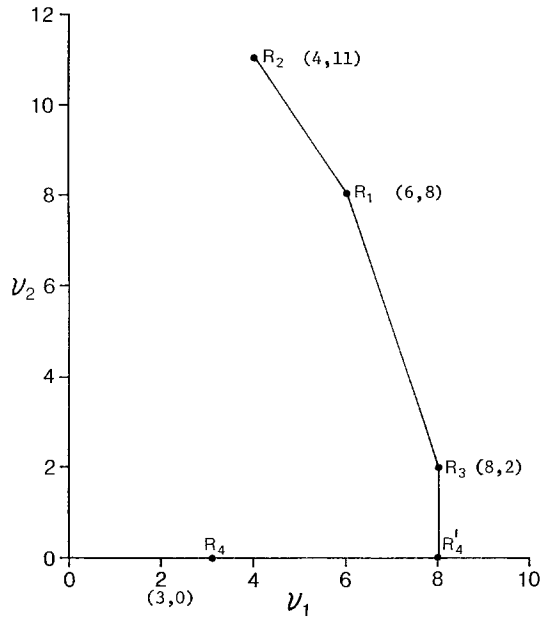

FIGURE 1. Desirability Frontier.

$R2$–$R1$–$R3$ as the *desirability frontier*, in that candidates i on this boundary have achieved the highest attainable index $Z_i = 1.0$. Any candidate behind the frontier (toward the origin) has a rating less than 1.0. In fact, the desirability rating for candidate #4 is given by the ratio of the line segment or range from the origin O to $R4$ to the line segment O to $R4'$. The desirability frontier therefore forms an *envelope* behind which all candidates lie, hence the name Data Envelopment Analysis.

Two important points must be emphasized here. First, different multiplier's (W_1, W_2) were used for the different candidates. Point A in Figure 2 defines the importance attached

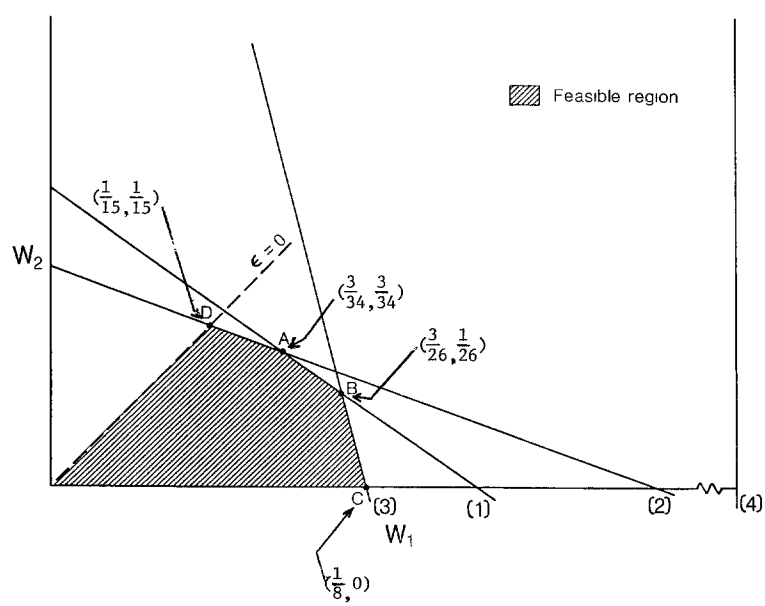

FIGURE 2. Feasible Weight Space for $\epsilon = 0$.

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to first and second place votes in evaluating the standing of candidate #2. Point B is used to evaluate candidate #3. Points A and B are both optimal for candidate #1, and C is optimal for candidate #4.

Second, if $d(j, \epsilon) = \epsilon$ and ϵ increases from its 0-level, the optimal points for certain candidates can change. Figure 2′ shows the shape of the feasible region when $\epsilon = \frac{1}{27}$. Here, the optimum for candidate #2, for example, is located at E. The resulting different coefficients of E versus A leads to a redefinition of the desirability frontier. See line segment R1–R2′ in Figure 1′. Now candidate #2 no longer has an index of $Z_2^* = 1$, meaning that the number of possible first place candidates has been reduced from three to two.

Clearly, the value of the discriminating factor ϵ influences the ranking of the candidates, in the sense that it discriminates among candidates on the frontier. In the section to follow we examine the impact of this factor.

3. A Model for Ranking the Candidates

In model (2.3), for any given value of ϵ one or more candidates i will achieve the maximum attainable desirability index $Z_i(\epsilon) = 1$. In this section we show that there is a maximum value for ϵ for which this problem has a solution, and demonstrate that the derivation of this value can lead to an ordinal ranking of the candidates. First a number of properties of (2.3) and $Z_i(\epsilon)$ are presented.

Property 3.1: For any i_0, $Z_{i_0}(\epsilon)$ is a monotonic nonincreasing function of ϵ.

PROOF: From the monotonicity of $d(j, \epsilon)$ it is clear that the feasibility region in (2, 3), for a given ϵ, contains the corresponding region for ϵ', if $\epsilon' > \epsilon$. Therefore, the objective function cannot increase. □

COROLLARY 3.1: *If* $Z_{i_0}(\epsilon_1) = 1$ *then* $Z_{i_0}(\epsilon) = 1$ *for all* $\epsilon \leq \epsilon_1$.

PROOF: This follows from Property 3.1 and the i_0-th constraint in (2.3). □

For a given function $d(j, \epsilon)$, define $I_1 = \{i; Z_i(0) = 1\}$. We assume that $d(j, \epsilon)$ is defined such that $I_1 \neq \varnothing$. For any $i \epsilon I_1$ let ϵ_i^* be the largest value of ϵ such that $Z_i(\epsilon) = 1$, and let $\epsilon_{\max}^* = \text{Max}_i\{\epsilon_i^*\}$. Clearly $\epsilon_{\max}^*$ may be ∞, for example when $d(j, \epsilon) = 0$ for all j and ϵ.

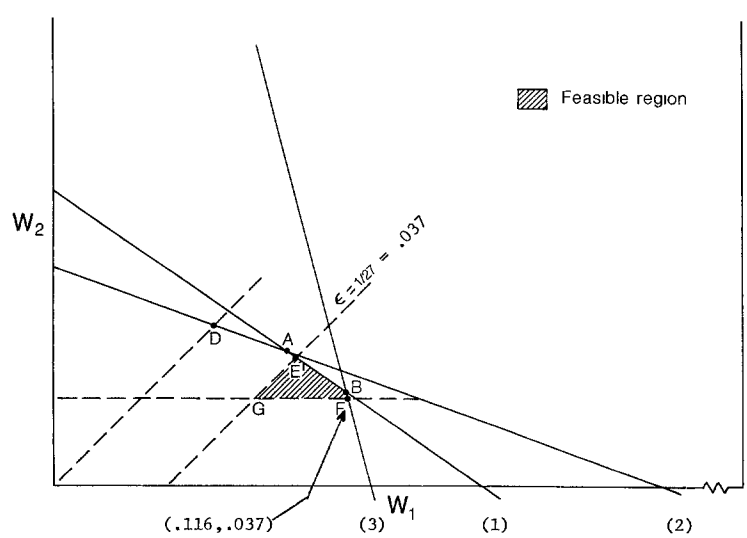

FIGURE 2′. Feasible Weight Space for $\epsilon = \frac{1}{27}$.

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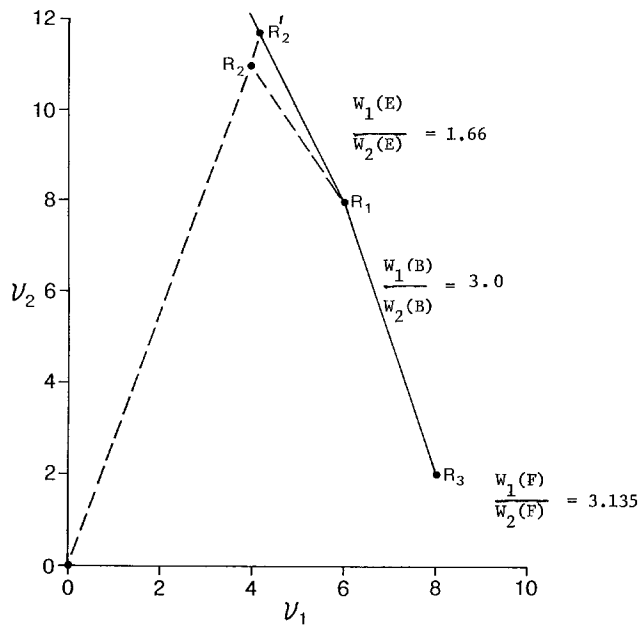

FIGURE 1′. Adjusted Desirability Frontier.

Property 3.2: $\epsilon^*_{\max}$ is the largest value of ϵ for which there exists a feasible solution for (2.3).

PROOF: Without loss of generality we can assume that $\epsilon^*_{\max} < \infty$. Let $\epsilon > \epsilon^*_{\max}$ and suppose there exists a feasible solution $W = (W_1, \ldots, W_k)$ for (2.3). Then, by the definition of $\epsilon^*_{\max}$ we must have that $Z_i(\epsilon) < 1$ *for all* $i = 1, \ldots, m$. The dual of (2.3) is

$$\begin{aligned} \text{Min} \sum_{i=1}^{m} x_i - \sum_{j=1}^{k} d(j, \epsilon) y_j & \\ \text{s.t.} \quad \sum_{i=1}^{m} x_i v_{i1} - y_1 & \geq v_{i_0 1}, \\ \sum_{i=1}^{m} x_i v_{ij} + y_{j-1} - y_j & \geq v_{i_0 j}, \quad j = 2 \cdots k, \\ x_i, y_i \geq 0. & \end{aligned} \tag{3.1}$$

From complementary slackness it follows that in optimality $x_1^* = \cdots = x_m^* = 0$. This implies that at least one of the constraints in (3.1) is violated, in contradiction. We conclude that the problem of determining the desirability index for each alternative applies only for ϵ in the closed interval $[0, \epsilon^*_{\max}]$, which is the *feasible range* of ϵ. □

THEOREM 3.1: *There exists an alternative* i_0 *such that* $Z_{i_0}(\epsilon) \geq Z_i(\epsilon)$, $\forall i = 1, \ldots, m$, $\forall \epsilon$ *in* $[0, \epsilon^*_{\max}]$.

PROOF: Let alternative i_0 be such that $\epsilon^*_{i_0} = \epsilon^*_{\max}$. Hence, $Z_{i_0}(\epsilon) = 1$ for all values of ϵ in the feasible range. Pick any alternative i such that $Z_i(\epsilon) = 1$ for $\epsilon \leq \epsilon_i^* \leq \epsilon^*_{\max}$, and $Z_i(\epsilon) < 1$ for $\epsilon_i^* < \epsilon \leq \epsilon^*_{\max}$. We conclude that there exists a nonempty set S of alternatives such that for *all* feasible values of the discriminating factors ϵ, the alternatives i_0 in S are superior, in terms of the desirability index, to all other alternatives. □

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Consider the problem

$$\begin{aligned} &\text{Max } \epsilon \\ &\text{s.t.} \quad \sum_{j=1}^{k} W_j v_{ij} \leq 1, \quad i = 1 \cdots m, \quad \text{(a)} \\ &W_j - W_{j+1} - d(j, \epsilon) \geq 0, \\ &W_k - d(k, \epsilon) \geq 0, \quad \text{(b)} \\ &W_i, \epsilon \geq 0. \end{aligned} \tag{3.2}$$

Property 3.3: At optimality, at least one of the constraints in (3.2a) holds at equality, that is there exists a candidate i_0 such that $\sum_{j=1}^{k} W_j v_{i_0 j} = 1$.

PROOF: Follows directly from the proof of Property 3.2. □

It follows from Property 3.3 and Theorem 3.2 that the solution to (3.2) yields all the winning candidate(s). Specifically, the candidates i for which $\sum_{j=1}^{k} W_j v_{ij} = 1$ at $\epsilon_{\max}$ dominate all other candidates in that they possess the highest level of *robustness* in terms of the range of ϵ. For the case $d(j, \epsilon) = \epsilon$, in a random sample of 60 problems, each with 20 voters and 10 candidates, 93% of the cases yielded a *single* first place candidate.

Having derived the first place candidate i_0, the i_0th constraint in (3.2a) can now be eliminated and (3.2) resolved. That candidate i_1 with $\sum W_j v_{i,j} = 1$ will hold second place, and the process is repeated.

The following theorem characterizes the nature of the W_j differentials vis-a-vis the discrimination intensity function.

THEOREM 3.2: *In optimality, all the constraints in* (3.2*b*) *hold as equalities*.

PROOF: Let $I_0 = \{i; \sum W_j^* v_{ij} = 1\}$ and without loss of generality assume that $v_{i1} > 0$ for all $i \in I_0$. Suppose that an optimal solution of (3.2) is such that $W_s - W_{s+1} - d(s, \epsilon) = \delta > 0$ for some s. Define

$$W'_j = \begin{cases} W_j - \delta, & j = 1 \cdots s, \\ W_j, & j = s+1, \ldots, k. \end{cases}$$

Clearly,

$$\sum_{j=1}^{k} W'_j v_{ij} < \sum_{j=1}^{k} W_j v_{ij} \leq 1 \qquad \forall i \in I_0,$$

$$\sum_{j=1}^{k} W'_j v_{ij} \leq \sum_{j=1}^{k} W_j v_{ij} < 1 \qquad \forall i \notin I_0,$$

$$W'_j - W'_{j+1} - d(j, \epsilon^*_{\max}) \geq 0, \qquad \text{with} \qquad W'_s - W'_{s+1} - d(s, \epsilon^*_{\max}) = 0.$$

It follows that $(W', \epsilon^*_{\max})$ is also an optimal solution for (3.2). For W', all the constraints in 3.2(a) are strict. But, according to Property 3.3 this is impossible. □

COROLLARY 3.2: *Problem* (3.2) *is equivalent to*

$$\begin{aligned} &\text{Max } \epsilon \\ &\text{s.t.} \quad \sum_{j=1}^{k} \Big(\sum_{l=j}^{k} d(l, \epsilon)\Big) v_{ij} \leq 1, \quad i = 1 \cdots m, \\ &\epsilon \geq 0. \end{aligned} \tag{3.3}$$

PROOF: From Theorem 3.2, the W_j can be recursively expressed as sums of the $d(l, \epsilon)$. □

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The ease with which problem (3.2) can be solved depends very much on the form of the $d(j, \epsilon)$ function. If d is a linear function, for example, (3.2) will be a linear programming problem. Such would be the case say where $d(j, \epsilon) = a_j\epsilon$, with $\{a_j\}$ being a set of scalars. Special cases of this would be: (i) a_j = constant (ii) $\{a_j\}$ is a monotonic increasing sequence (larger gaps between W_j & W_{j+1} as j increases); (iii) $\{a_j\}$ is a decreasing sequence (smaller gaps as j increases).

An important subclass of discrimination intensity functions is that for which $d(j, \epsilon) = g(j)h(\epsilon)$, where $h(\epsilon)$ is strictly monotonic increasing in ϵ. The linear examples of the previous paragraph fall into this category. For this subclass, problems (3.2) and (3.3) have closed form solutions. Specifically, problem (3.3) becomes:

$$\text{Max } \epsilon$$

$$\text{Subject to} \qquad h(\epsilon) \sum_{j=1}^{k} \Big(\sum_{l=j}^{k} g(l)\Big) v_{ij} \le 1, \qquad \epsilon \ge 0, \qquad \text{and}$$

$$\epsilon^*_{\max} = \min_i h^{-1}\Big[\Big(\sum_{j=1}^{k} \Big(\sum_{l=j}^{k} g(l)\Big) v_{ij}\Big)^{-1}\Big]. \tag{3.4}$$

A winning candidate i is, therefore, the one for which this minimum is attained. Due to the monotonicity of h, it is sufficient to find the candidate i for which

$$\sum_{j=1}^{k} \sum_{l=j}^{k} (g(l)) v_{ij} \tag{3.5}$$

is maximized.

Special Cases: (1) Exponentially decreasing intensity of discrimination values: $g(j) = e^{-\alpha j}$.

The maximum of (3.5) then becomes:

$$\max_i \sum_{j=1}^{k} \left(\frac{1 - e^{-\alpha(k-j+1)}}{e^{\alpha j} - 1}\right) v_{ij}.$$

The candidate i for which the above maximum is attained is the winner.

(2) Constant intensity of discrimination values: $g(j) = 1$. In this case, the winner is a candidate with the maximum value of $\sum_{j=1}^{k} (k - j + 1) v_{ij}$.

This is, of course, the well-known Borda method of marks. Consequently, Borda's (and Kendall's 1962) model for deriving a consensus among a set of voter rankings is a special case of our model when $g(j)$ = a constant.

EXAMPLE: Consider the case of 20 voters, each of whom is asked to rank 4 out of 6 candidates on a ballot. Let the outcome from the vote be as shown in matrix V.

$V =$

	Standing			
	1	2	3	4
a	3	3	4	3
b	4	5	5	2
c	6	2	3	2
d	6	2	2	6
e	0	4	3	4
f	1	4	3	3

For example, candidate "a" receives 3 first, 3 second, 4 third and 3 fourth place votes. i.e. $v_{a1} = 3$, $v_{a2} = 3$, $v_{a3} = 4$, $v_{a4} = 3$.

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Case 1: $d(j, \epsilon) = \epsilon \Rightarrow b$ is the winner, $\epsilon^*_{\max} = 0.0233$. (This is the Borda/Kendall winner.)

Case 2: $d(j, \epsilon) = \epsilon/j \Rightarrow d$ is the winner, $\epsilon^*_{\max} = 0.0577$.

Case 3: $d(j, \epsilon) = \epsilon/j! \Rightarrow c$ is the winner, $\epsilon^*_{\max} = 0.0808$.

In Case 1 the discrimination intensity is evenly distributed among the rank positions, therefore the fact that candidate b had the largest number of votes for rank positions 1, 2 and 3 played a major role in determining him as the winner. In Case 3 on the other hand, the discrimination intensity function is exponentially decreasing in j, which implies that the first place vote has a relatively very large weight while the fourth position has very little weight in determining the winner. Although candidate d received more total votes (16) than candidate c (13), the fact that d's advantage was concentrated in the fourth position while in the third position c had a slight advantage on d, (in positions 1 and 2 they had identical score) plays a key role in determining c as a winner for the $d(j, \epsilon) = \epsilon/j!$ case.

4. Discussion

The problem of aggregating votes in a preferential election has been a subject of study for over 200 years. In this paper a model is presented which aggregates votes into an overall index in a way that allows each candidate to be assessed in a fair manner. The model amounts to determining for each candidate i the best set of weights W_j to apply to jth place standings v_{ij} for that candidate. We define a discrimination intensity function $d(j, \epsilon)$ which specifies the minimum amount by which the multipliers W_j and W_{j+1} must differ. It is shown that for a certain general subclass of these functions the winning candidate can be obtained by a closed form expression. Furthermore, it is shown that for the special case in which all consecutive pairs of weights deviate by the same amount ($d(k, \epsilon) = d(j, \epsilon)$), our model is equivalent to the well-known models of Borda and Kendall.[1]

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